

14. If the absolute refractive indices of two media X and Y are $\frac{6}{5}$ and $\frac{4}{3}$ respectively, then the refractive index of Y with respect to X will be :

(A) $\frac{10}{9}$

(B) $\frac{9}{10}$

(C) $\frac{9}{8}$

(D) $\frac{8}{9}$

15. An object is placed at a distance of 30 cm from the pole of a concave mirror. If its real and inverted image is formed at 60 cm in front of the mirror, the focal length of the mirror is :

(A) -15 cm

(B) -20 cm

(C) $+20$ cm

(D) $+15$ cm

- 25.** (a) Out of the two lenses, one concave and the other convex, state which one will diverge a parallel beam of light falling on it. Draw a ray diagram to show the principal focus of the lens. 2

OR

- (b) A ray of light after refraction from a convex lens emerges parallel to its principal axis. 2
- (i) Draw a labelled ray diagram to show it.
- (ii) In this case, the incident ray before refraction from the lens passes through a point on its principal axis. Name the point.

31. A convex mirror used for rear view on an automobile has a focal length of 1.5 m. If a 3 m high bus is located at 6.0 m from the mirror, use mirror formula to determine the position and size of the image of the bus as seen in the mirror.

- 31.** A convex lens forms an 8.0 cm long image of a 2.0 cm long object which is kept at a distance of 6.0 cm from the optical centre of the lens. If the object and the image are on the same side of the lens, find (i) the nature of the image, (ii) the position of the image, and (iii) the focal length of the lens.

13. Mirror 'X' is used to concentrate sunlight in solar furnace and Mirror 'Y' is fitted on the side of the vehicle to see the traffic behind the driver. Which of the following statements are true for the two mirrors ?

- (i) The image formed by mirror 'X' is real, diminished and at its focus.
 - (ii) The image formed by mirror 'Y' is virtual, diminished and erect.
 - (iii) The image formed by mirror 'X' is virtual, diminished and erect.
 - (iv) The image formed by mirror 'Y' is real, diminished and at its focus.
- (A) (i) and (ii) (B) (ii) and (iii)
- (C) (iii) and (iv) (D) (i) and (iv)

24. An object is placed at a distance of 60 cm from a concave lens of focal length 30 cm. Use lens formula to find the position of the image formed in this case.

31. Draw ray diagrams to show the nature, position and relative size of the image formed by a convex mirror when the object is placed (i) at infinity and (ii) between infinity and pole P of the mirror.

- 36.** (a) (i) The power of a lens 'X' is -2.5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens ?
- (ii) "The value of magnification 'm' for a lens is -2 ." Using new Cartesian Sign Convention and considering that an object is placed at a distance of 20 cm from the optical centre of this lens, state :
- (I) the nature of the image formed;
 - (II) size of the image compared to the size of the object;
 - (III) position of the image, and
 - (IV) sign of the height of the image.
- (iii) The numerical values of the focal lengths of two lenses A and B are 10 cm and 20 cm respectively. Which one of the two will show higher degree of convergence/divergence ? Give reason to justify your answer.

OR

- (b) (i) Draw a ray diagram to show the refraction of a ray of light through a rectangular glass slab when it falls obliquely from air into glass.
- (ii) State Snell's law of refraction of light.
- (iii) Differentiate between the virtual images formed by a convex lens and a concave lens on the basis of :
 - (I) object distance, and
 - (II) magnification.

21. An object is placed at a distance of 10 cm in front of a concave mirror of focal length 15 cm. Use mirror formula to determine the position of the image formed by this mirror.

11. An optical device 'X' is placed obliquely in the path of a narrow parallel beam of light. If the emergent beam gets displaced laterally, the device 'X' is :

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- (a) plane mirror
- (b) convex lens
- (c) glass slab
- (d) glass prism

13. Absolute refractive index of water and glass is $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is :

(A) $\frac{9}{4}$ m/s

(B) $\frac{7}{3}$ m/s

(C) $\frac{16}{9}$ m/s

(D) $\frac{9}{8}$ m/s

- 34.** (a) (i) “In refraction of light through a rectangular glass slab, the emergent ray is always parallel to the direction of the incident ray.” Why ? Explain with the help of a ray diagram. What happens when a ray of light falls normally on one of the faces of a rectangular glass prism ? Draw diagram.
- (ii) An object is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 20 cm. Use Lens formula to determine the position of the image formed in this case.

- (b) (i) A student wishes to study the image formation by a concave mirror using candle flame as object. State the type of the image formed by the mirror and mention the change in the image formed, if any, that he observes when the candle flame is gradually moved away from the pole of the mirror. Draw a ray diagram to show the image formation when the object distance is nearly equal to the radius of curvature of the mirror.
- (ii) A convex mirror used for rear-view on an automobile has a focal length of 3.0 m. If a bus is located at 6.0 m from this mirror, use mirror formula to find the position of the image of the bus as seen in the mirror.

16. To get an image of magnification -1 on a screen using a lens of focal length 20 cm, the object distance must be :

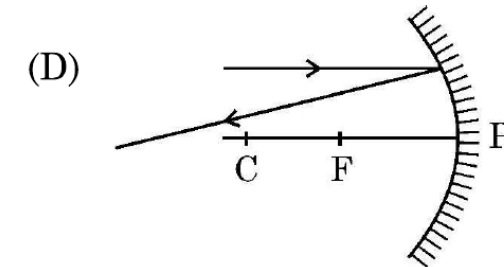
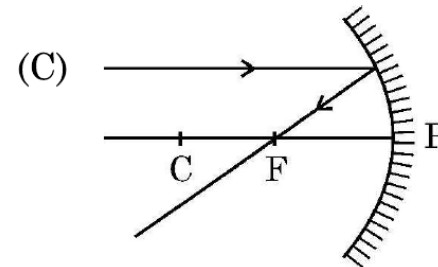
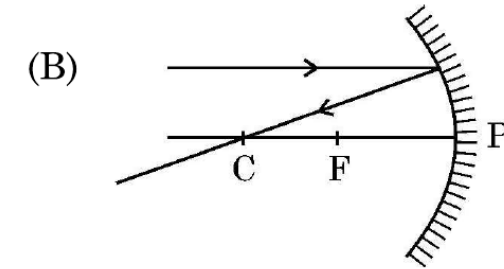
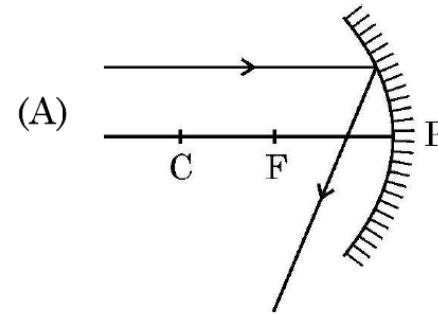
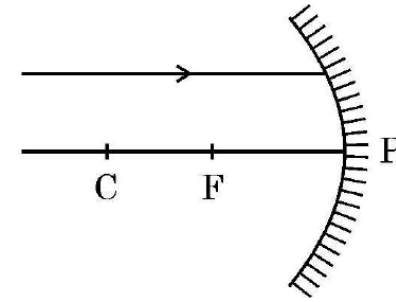
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- (a) Less than 20 cm
- (b) 30 cm
- (c) 40 cm
- (d) 80 cm

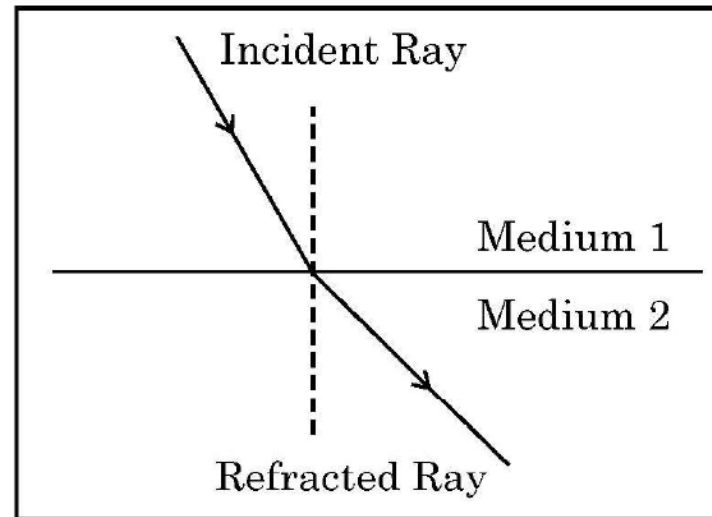
27. If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer.

14. Identify from the following the ray diagram which shows the correct path of the reflected ray for the ray incident on a concave mirror as shown :

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22. Study the figure in which the path of a ray of light going from Medium 1 to Medium 2 is shown.



- (a) Out of the two Media – Medium 1 and Medium 2, in which is the speed of light more ?
- (b) State reason of bending of the refracted ray away from the normal.
- (c) Express refractive index of Medium 2 with respect to Medium 1 in terms of speed of light in two media.

29. A student placed a candle flame at different distances from a convex lens and focused its image on a screen. He recorded his observation in tabular form as given below :

S.No.	Distance of flame from the lens (cm)	Distance of the image from the lens
1	– 90	+ 18
2	– 60	+ 20
3	– 40	+ 24
4	– 30	+ 30
5	– 24	+ 40
6	– 20	+ 60
7	– 18	+ 90
8	– 12	+ 120

Analyse the observation table and on the basis of your analysis only, answer the following questions (without doing any calculations) :

- (a) What is the focal length of the convex lens used ? Give reason to justify your answer.
- (b) Which one of the sets of observations is not correct and why ?
- (c) Draw ray diagram to show image formation for any correct set of observation.

12. An object is placed at a distance of 30 cm from the reflecting surface of a concave mirror of radius of curvature 40 cm. The image formed is

- | | |
|---------------------------|----------------------------|
| (A) Virtual and magnified | (B) Virtual and diminished |
| (C) Real and magnified | (D) Real and diminished |

31. An object of height 4.0 cm is placed at a distance of 30 cm from the optical centre of a convex lens of focal length 20 cm. Use lens formula to determine (i) the distance of the image from the optical centre and (ii) height of the image formed.

16. In order to obtain large images of the teeth of patients, the dentist holds the concave mirror in such a manner that the teeth are positioned
- (A) at the focus of mirror.
 - (B) between pole and focus of the mirror.
 - (C) between focus and centre of curvature of the mirror.
 - (D) at the centre of curvature of the mirror.

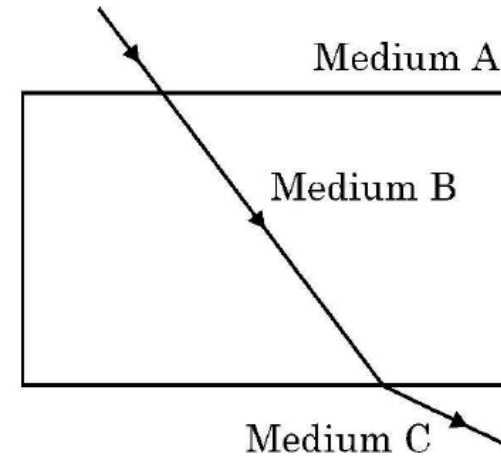
21. During an experiment, a student observes that the given mirror is forming an erect and diminished image for all positions of the object placed in front of it.

Based on the this information, answer the following questions :

- (a) Identify the mirror.
- (b) Using new Cartesian sign convention, write the magnitude of the magnification in this case.
- (c) What will be the radius of curvature of the mirror if its focal length is 20 cm ?

29. The power of a lens is -0.25 D. Based on this information, find out
- (a) The type of lens and its focal length.
 - (b) The eye defect for which it may be used as a corrective lens.
 - (c) The nature and size of the image formed by this lens when an object is placed between F and $2F$ from the optical centre of this lens.

36. (A) (a) Observe the following diagram and compare (i) speed of light and (ii) optical densities of the three media A, B and C. Also give justification for your answer of any one of the two cases in terms of refractive indices of A, B and C.



- (b) Redraw the path of a ray of light through the three media, if the ray of light starting from medium A falls on the medium B
- (i) Obliquely and the optical density of medium B is made more than that of A and C.
 - (ii) The ray falls normally from medium A to medium B.

- (B) Analyse the following observation table showing variation of image distance (v) with object distance (u) in case of a convex lens and answer the questions that follow without doing any calculations :

Observation Number	Object distance (u) in cm	Image distance (v) in cm
1	-15	-60
2	-25	+100
3	-30	+60
4	-40	+40
5	-60	+30
6	-100	+25

- (a) Determine the focal length of the lens. Give reason for your answer.
- (b) Find magnification of the image formed in Observation No. 3.
- (c) The numerical value of magnifications in cases of observation 1 and 2 is same. List two differences in the images formed in these two cases.

24. The values of absolute refractive indices of kerosene and water are 1.44 and 1.33 respectively. Compare the two media on the basis of their

- (a) optical density
- (b) mass density
- (c) relative speed of propagation of light.

What do you infer on the basis of above comparisons ?

23. When an incident ray of light enters a medium 'X' from air, it bends towards the normal.

Infer about the following with respect to air :

- (a) Optical density of medium 'X'.
- (b) Speed of light through medium 'X'

Give reason for your answer in each case.